

How AI and AGI Can Change the World for Good

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Abstract

Most public discussion of artificial intelligence focuses on its dangers. This paper looks at the other side: how today's AI and a future artificial general intelligence (AGI) could improve human life. We review evidence that AI is already helping in health, science, education, climate, work and access to information, and we describe what a general system could add on top of this. We argue that the benefits are real and potentially very large, but not automatic. They depend on safety, fair access and good decisions by the people who build and govern AI. We also describe how small teams, including those serving under-represented communities, can take part.

1. Introduction

Every major technology has changed how people live: the printing press spread knowledge, electricity powered industry, and the internet connected the world. AI may be different in one important way. Earlier technologies extended human muscle or communication. AI extends human thinking.

This paper asks three questions:

- 1 Where is today's narrow AI already doing good?
- 2 What more could AGI do, if it is built safely?
- 3 What must be true for these benefits to reach everyone, not only a few?

We separate what has already happened, which we support with published examples, from what is possible in the future, which we mark clearly as potential.

2. Health and medicine

Health is the area where AI's positive impact is clearest so far.

Area	What AI does	Example
Protein structure	Predicts the 3D shape of proteins from their sequence, a task that once took years per protein	AlphaFold predicted structures for over 200 million proteins (Jumper et al., 2021). Its creators shared the 2024 Nobel Prize in Chemistry
Medical imaging	Finds signs of disease in scans, X-rays and eye images	AI systems have matched specialists in some screening tasks, such as detecting diabetic eye disease (Gulshan et al., 2016)
Drug discovery	Searches huge numbers of possible molecules to find promising medicines faster	An AI model helped identify a new antibiotic, halicin (Stokes et al., 2020)
Access to care	Answers health questions and helps triage patients where doctors are scarce	Useful in regions with few doctors, although it must be checked for errors

What AGI could add. A general system could read all medical research, design experiments, suggest treatments for rare diseases and help doctors personalise care. Some researchers argue that AI could compress decades of biological research into a few years (Amodei, 2024). This is a possibility, not a promise.

3. Science and discovery

AI is becoming a tool for doing science itself, not only for analysing results.

- **Materials.** An AI system proposed about 2.2 million new crystal structures, including hundreds of thousands predicted to be stable, which could lead to better batteries and solar cells (Merchant et al., 2023).
- **Fusion energy.** Reinforcement learning has been used to control the plasma inside a fusion reactor (Degraeve et al., 2022).
- **Mathematics.** AI systems now help find proofs and solve competition-level maths problems.
- **Weather.** AI weather models can produce accurate 10-day forecasts in minutes on a single machine (Lam et al., 2023).

What AGI could add. Today, humans ask the questions and AI helps answer them. An AGI could form its own hypotheses, design and run experiments with robotic labs, and learn from the results. If this works, the speed of discovery itself could increase.

4. Education

Research has long shown that one-to-one tutoring greatly improves learning. In a famous study, students with a personal tutor performed about two standard deviations better than students in a normal class (Bloom, 1984). The problem was that personal tutors were too expensive for most people.

AI tutors can make personal tutoring affordable. They can explain a topic in many ways, adapt to each student's level, answer questions at any hour and work in many languages.

Benefit	Why it matters
Personal pace	Fast learners are not held back; slower learners are not left behind
Always available	Students can learn at night, on weekends or while working
Any language	Learners can study in their own language, not only English
Self-taught learners	People without access to universities can still learn advanced subjects

The author of this paper is himself self-taught in programming and many languages. AI tools make this path easier for millions of others.

5. Language and inclusion

Around 7,000 languages are spoken today, but most technology works well in only a few. This leaves many communities without equal access to information, services and opportunity.

- Machine translation projects now cover around 200 languages, including many with few speakers online (NLLB Team, 2022).
- Speech recognition and text-to-speech help people who cannot read or write, and people with visual impairments.
- Small teams can now build models for their own communities. For example, the Romani language, spoken by millions across Europe, has very little digital support. Projects at Amaro Than Platforms, such as a Romani-language AI model, aim to change this.

AI that works in every language could help protect endangered languages and give minority communities a voice in the digital world.

6. Climate and environment

- **Better forecasts** help farmers plan and help communities prepare for storms, floods and heat waves.
- **Energy grids** can balance supply and demand more efficiently, making it easier to use solar and wind power.
- **New materials** discovered with AI could improve batteries, solar panels and carbon capture.
- **Monitoring** with satellite images can track deforestation, pollution and wildlife.

AI also uses a lot of energy itself. For AI to help the climate overall, data centres must become more efficient and use clean energy.

7. Work and the economy

Early studies suggest AI can make workers more productive, especially less experienced ones.

- In one large study, customer support agents using an AI assistant resolved about 14% more issues per hour on average, with the biggest gains for newer workers (Brynjolfsson et al., 2023).
- In an experiment on writing tasks, people using ChatGPT finished faster and produced higher-quality work (Noy and Zhang, 2023).
- AI coding assistants help developers write and fix code faster, and allow small teams to build what once needed large companies.

What AGI could add. If AGI can do most intellectual work, it could produce enormous economic growth and make goods and services much cheaper. This could reduce poverty. But it could also displace many jobs. Whether the result is shared wealth or deeper inequality depends on policy choices, such as education, retraining and how the gains are distributed.

8. From AI to AGI: what changes

Today's AI (narrow)	AGI (potential)
Helps humans with specific tasks	Could carry out whole projects from start to finish
Needs humans to ask the right questions	Could find new questions and problems on its own
Learns mainly during training	Could learn continuously from experience
Works in one domain at a time	Could connect ideas across medicine, physics, economics and more
Speeds up individual experts	Could act like millions of expert researchers working in parallel

A good AGI future might look like this: diseases that are cured rather than managed, clean and cheap energy, a personal teacher and doctor for every person, and scientific progress that once took a century happening in a decade. These are the hopes that motivate many AI researchers, including the author.

9. Conditions for a good outcome

None of these benefits happen automatically. We identify five conditions:

Condition	What it means
Safety	AI systems must be reliable, honest and aligned with human values. A powerful system that makes mistakes or pursues the wrong goals can cause great harm
Fair access	Benefits must reach poor countries, small languages and minority communities, not only rich companies and countries
Human control	People must stay in charge of important decisions, especially in health, law and government
Education	People need the skills to use AI well and to adapt as jobs change
Good governance	Clear rules, transparency and international cooperation, so that AI serves the public interest

Safety and benefit are not opposites. The safer and more trusted AI is, the more widely it can be used for good.

10. What small teams can do

Large labs build the most powerful models, but small teams have an important role:

- **Serve communities big companies overlook**, such as speakers of minority languages.
- **Fine-tune open models** for local needs in health, education and public services.
- **Test and audit models** so that they are safe and reliable for their users. Tools such as Auditor (Berisha, 2026) help small teams check their own models.
- **Share work openly**, so others can build on it.

11. Conclusion

AI is already doing real good: predicting protein structures, finding new medicines, improving forecasts, helping students and making workers more productive. AGI, if it is built safely, could multiply these benefits and help solve problems that have troubled humanity for centuries.

The future of AI is not decided by the technology alone. It will be decided by the people who build it, the rules societies set and whether its benefits are shared. If we get these choices right, AI and AGI could be among the most positive changes in human history.

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